QUANT

Time, Speed & Distance

Assignment 04

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Time, Speed & Distance Online Assignment -04

1. What is the average speed if a man drives 3 hours at 60 km/hr and the next 6 hours at 50 km/hr.?

1. 55 km/hr 2. 52.5 km/hr 3. 53.33 km/hr 4. 56.67 km/hr

2. Train A leaves from P to Q. Half an hour later train B leaves for P from Q. Two hours after A had started, the trains meet. A has covered 50 % more distance than train B when they meet. What is the ratio of the speeds of train A to train B?

1. 6 : 5 2. 5 : 6 3. 9 : 8 4. 8 : 9

3. A and B run around a circular park of circumference 1 km starting simultaneously from the same point and in the same direction at 4 km/hr and 6 km/hr. When they meet again for the first time at the starting point, how many laps would A have completed?

1. 3 2. 2 3. 4 4. 6

4. A and B run around a circular path of circumference 880 yards, starting at the same point in the same direction simultaneously. A runs at a speed of 3 mph and B at 5 mph. How many points on the circular path exist relative to the starting point in which they will meet every time they are together? (1 mile = 1760 yards)

1. 2 2. 4 3. Starting point 4. Infinite points

5. Two trains leave New Delhi Railway Station. The first train travels due west and the second train due north. The first train travels 5 km/hr faster than the second train. If after two hours they are 50 km apart, find the average speed of each train (in kmph).

1. 15, 10 2. 10, 5 3. 25, 20 4. 20, 15

6. How long will it take to row 20 km upstream if one can row 10 km in 10 minutes in still water and the same distance in 8 minutes with the stream?

1. 12 min 2. 13.33 min 3. 24 min 4. 26.67 min

7. I cover $^{2}/_{3}^{rd}$ of the distance that I have to travel at 40 km/hr and the remaining at 30 km/hr. What is the average speed at which I cover the entire distance?

1. 34.29 km/hr 2. 35 km/hr 3. 32.72 km/hr 4. 36 km/hr

8. A train travelling at 78 km/hr crosses a girl sitting in a train of length 110 m travelling in the same direction at 42 km/hr in 20 seconds. The length of the faster train is

1. 90 meter 2. 110 meter 3. 200 meter 4. 100 meter

9. A boat makes a return journey from point A to point B and back in 5 hours 36 minutes. One way it travels with the stream and on the return it travels against the stream. If the speed of the stream increases by 2 km/hr, the return journey takes 9 hours 20 minutes. What is the speed of the boat in still water? (The distance between A and B is 16 km.)

1. 5 km/hr 2. 3 km/hr 3. 7 km/hr 4. 9 km/hr

10. Two places P and Q are 162 km apart. A train leaves P for Q and at the same time another train leaves Q for P. The two trains meet after 6 hours. If the train travelling from P to Q travels 8 km an hour faster than the other, find the speed (in kmph) of the two trains.

1. 17.5, 9.5 2. 16.5, 8.5 3. 17.5, 10.5 4. 10, 16

Answer key & Explanations

Q No.	Key	Explanation
1.	3	Average speed = $\frac{\text{Total distance}}{\text{Total time}} \Rightarrow \frac{(3\times 60)+(6\times 50)}{3+6} = \frac{480}{9} = 53.33$ km/hr.
2.	3	Let speed of A is x and speed of B is y. $\therefore$ Distance covered by A in 2 hrs = $2x$ and distance covered by B in 1.5 hrs = $1.5y$. Also $2x = 1.5(1.5y)$ $\Rightarrow x : y = 9 : 8$.
3.	2	Time taken by A to cover the whole circumference = $\frac{1}{4}$ hrs and for B, time taken = $\frac{1}{4}$ hrs. Hence they would be together after LCM of $\frac{1}{4}$ *and* $\frac{1}{6}$ i.e. $\frac{1}{2}$ *hrs.* So at the meeting point, A would have completed $\frac{1}{2}\times 4 = 2$ laps.
4.	1	Total circumference = $\frac{1}{2}$ mile $\therefore$ time taken by A for complete circumference = $\frac{1}{6}$ hrs and similarly time taken by B = $\frac{1}{10}$ *hrs.* Hence they would be together after LCM of $\frac{1}{6}$ and $\frac{1}{10}$ *hrs.* $\frac{1}{2}$ hrs. So there will be 2 points in which they will meet every time. i.e. at starting point and at mid-point of the circumference.
5.	4	Let the speeds of each train be $x + 5$ and x km / hr. $\therefore$ as one is in West direction and other in North direction $\therefore$ they will form a right–angled triangle. After 2 hrs, distance covered by trains is $2x$ and $2x + 10$km. Their shortest distance = 50 km. $\therefore (2x + 10)^2 + (2x)^2 = (50)^2$. Solving this equation, we get x = 15 km/hr and hence speed of faster train = 15 + 5 = 20 km/hr.
6.	4	Let x be the speed of man in still water and y be the speed of stream. $\therefore$ Speed of man (x) = 60 km/hr and speed of stream = 75 km/hr. (downstream) $\therefore$ Speed of stream = 15 km/hr. Hence upstream speed = 60 – 15 = 45 km/hr. So time taken to cover 20 km = $\frac{20}{45}\times 60 = 26.67$ min.
7.	4	Average speed = $\frac{\text{Total distance}}{\text{Total time}}$. Let the total distance = D. $\therefore$ Average speed = $\frac{D}{\left(\frac{2D}{3}\times\frac{1}{40}\right)+\left(\frac{D}{3}\times\frac{1}{30}\right)} = \frac{D}{\frac{D}{60}+\frac{D}{90}} \Rightarrow \frac{D\times 180}{5D} \Rightarrow 36 km/hr.$
8.	3	Let the length of the faster train = x. $\therefore \frac{(x)18}{(78-42)5} = 20 \Rightarrow x = 200$ meters.
9.	3	Let x be speed of u / s and y be the speed of d / s. $\therefore \frac{16}{x}+\frac{16}{y} = \frac{28}{5}$ and $\frac{16}{y+2}+\frac{16}{x-2} = \frac{28}{3}$. Solving these 2 equations, we get x = 4km/hr and y = 10km/hr $\therefore$ speed of boat in still water = $\frac{4+10}{2} = 7 km/hr.$

10.	1	Time taken to meet = 6 hours. Let the 2 speeds be x km / hr and y km / hr. Total distance = 162 km. Difference of speeds = 8km / hr. $\therefore x - y = 8$km / hr. These 2 trains are in opposite directions $\because$ relative speed = $x + y = 162 / 6 = 27$ km / hr $\Rightarrow x = 17.5$ km / hr and $y = 9.5$ km / hr.